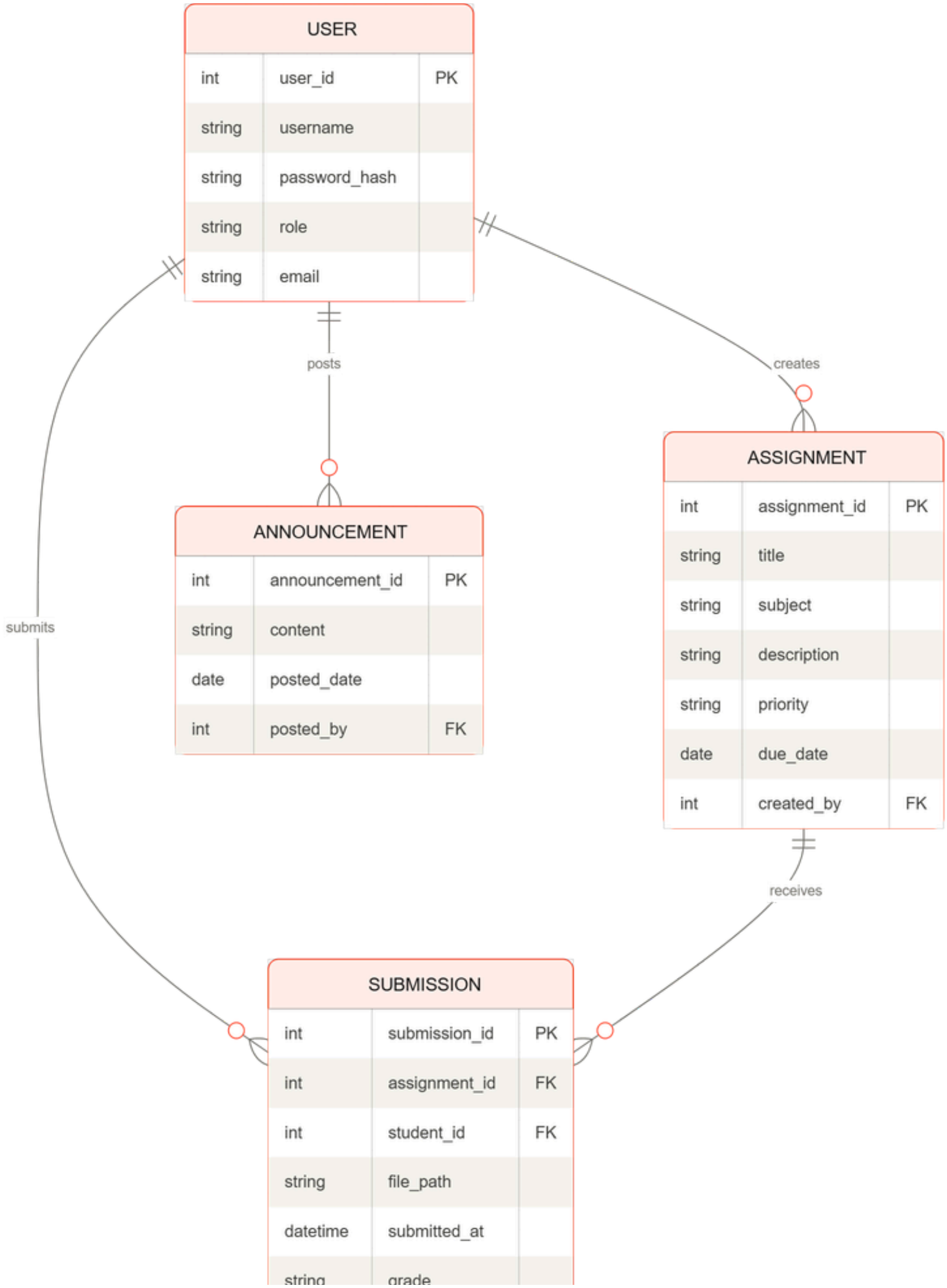
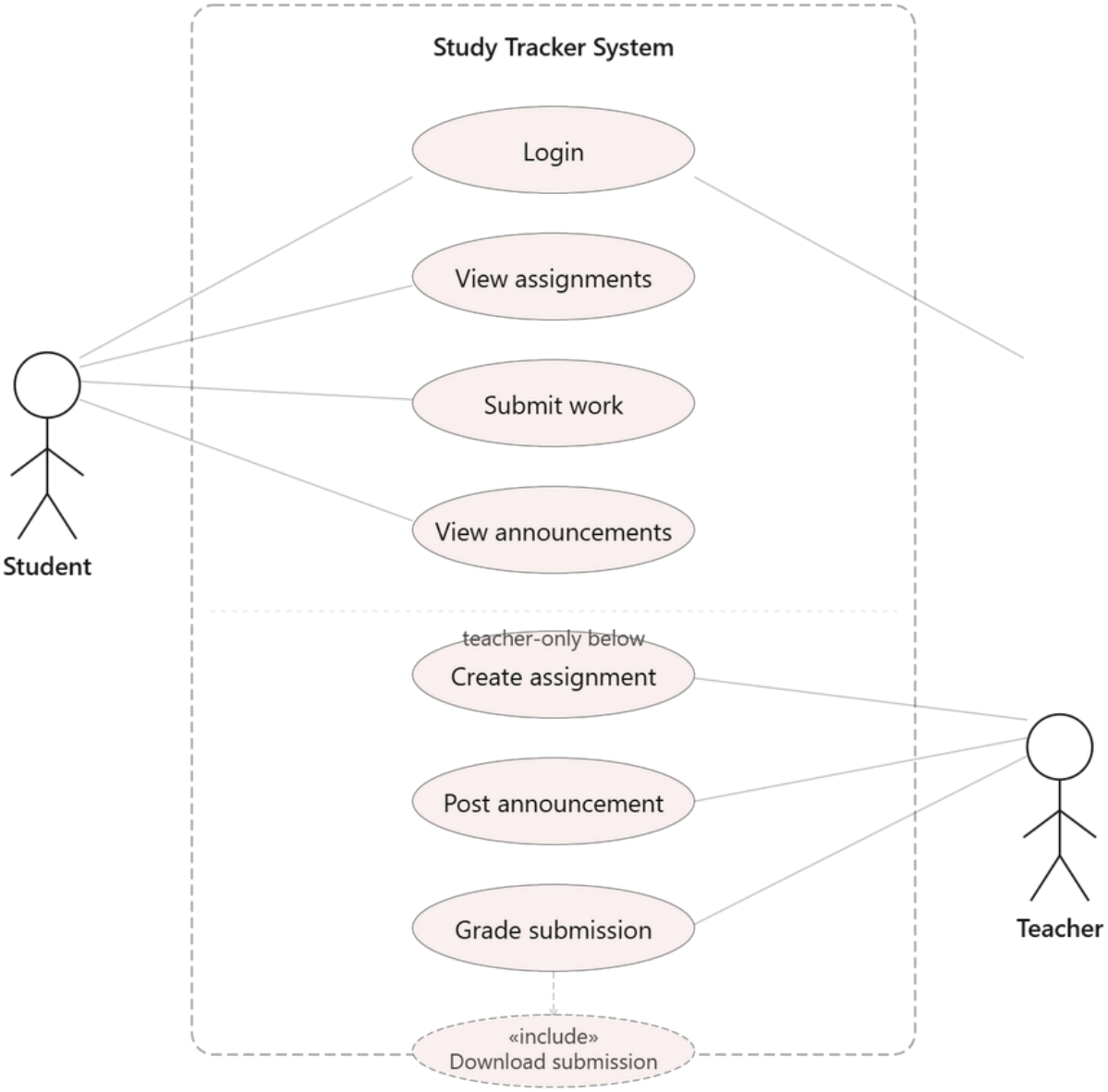

USER
int user_id PK
string username
string password_hash
string role
string email
posts
creates
ANNOUNCEMENT
int announcement_id PK
string content
date posted_date
int posted_by FK
ASSIGNMENT
int assignment_id PK
string title
string subject
string description
string priority
date due_date
int created_by FK
submits
receives
SUBMISSION
int submission_id PK
int assignment_id FK
int student_id FK
string file_path
datetime submitted_at
string grade

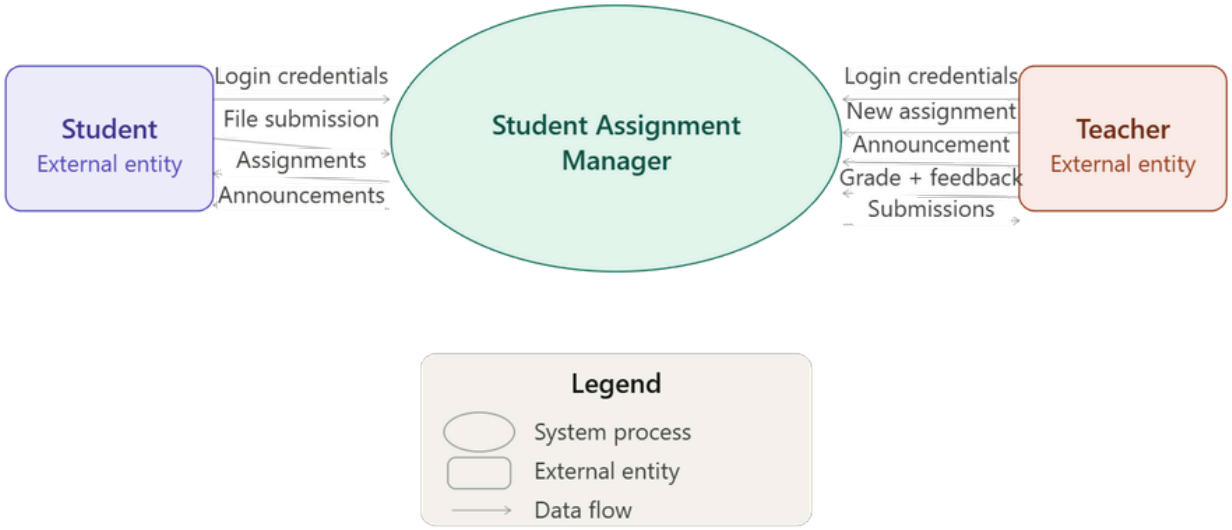
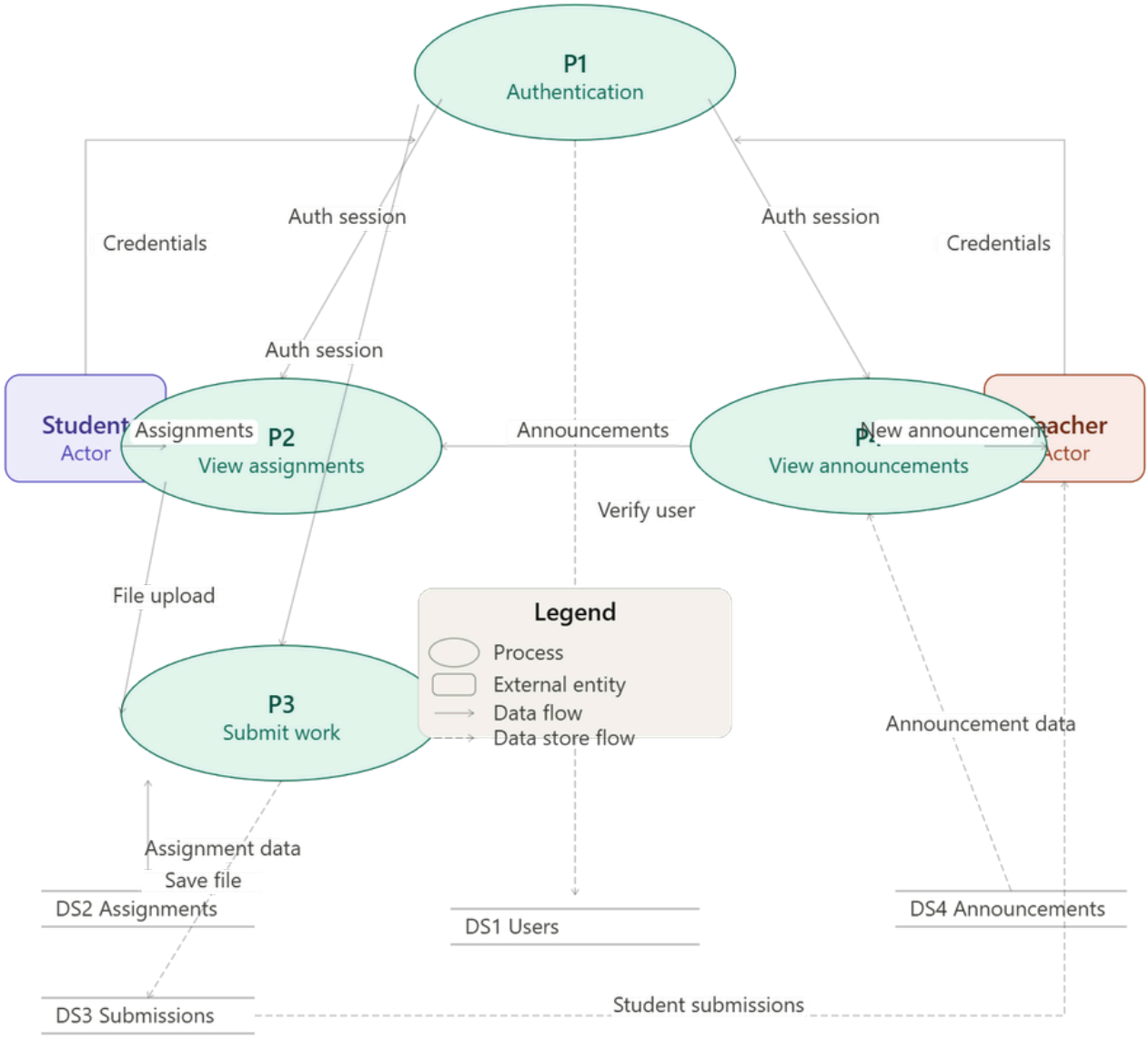
All entities | User | Assignment | Submission | Announcement

Field	Type	Key	Required	Description
USER				
user_id	INT	PK	Required	Unique identifier for each user (auto-increment)
username	VARCHAR(50)		Required	Unique login name; used on the login screen
password_hash	VARCHAR(255)		Required	Bcrypt-hashed password; never stored in plain text
role	ENUM('student','teacher')		Required	Determines dashboard view — 'student' or 'teacher'
email	VARCHAR(100)		Optional	Optional email address for notifications
ASSIGNMENT				
assignment_id	INT	PK	Required	Unique identifier for each assignment (auto-increment)
title	VARCHAR(100)		Required	Short assignment name shown on student dashboard
subject	ENUM('Math','English','Science',...)		Required	Subject area; drives the subject tag shown in the UI
description	TEXT		Optional	Detailed instructions shown to the student (what to do)
priority	ENUM('Low','Medium','High')		Required	Importance level; shown as slider on teacher form
due_date	DATE		Required	Deadline displayed under each assignment card
created_by	INT	FK	Required	References the teacher who posted the assignment
SUBMISSION				
submission_id	INT	PK	Required	Unique identifier for each file submission (auto-increment)
assignment_id	INT	FK	Required	Links submission to its parent assignment
student_id	INT	FK	Required	Links submission to the student who uploaded it
file_path	VARCHAR(255)		Required	Server path to the uploaded file (drag-and-drop / browse)
submitted_at	DATETIME		Required	Timestamp set automatically when file is uploaded
grade	ENUM('A','B','C','D','F')		Optional	Grade assigned by the teacher; null until graded



Use Case Diagram

Context diagram — DFD level 0



DFD level 1 — Student assignment manager